

Week # 9
Payback (PB) Period and Service - Life Analysis

Payback Period

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What is the Payback Period?

The payback period refers to the amount of time it takes to recover the cost of an investment.

Simply the payback period is the length of time an investment reaches a break-even point

The desirability of an investment is directly related to its payback period.

Shorter paybacks mean more attractive investments.

Although calculating the payback period is useful in financial and capital budgeting has applications in other industries.

It can be used by homeowners and businesses to calculate the return on energy-efficient technologies such as solar panels and insulation, including maintenance and upgrades.

Details of Payback Period

1. PB time period is the period in which investment is recovered through cash – inflow.

2. From investor point of view PB period should be as short as possible to ensure investment is not held up too long.

There are 3 types methods to determine PB period depending upon cash inflow:

1. PB period for even cash flow
2. PB period for uneven cash flow
3. PB period for Discounted Cash Flow -> * time – value money is considered
* more reliable
* covers risk

1) PB period for even cash inflow

In PB period for even cash inflow the cash inflow is uniform over the period like \$ 100 / year etc.

$$PB = \frac{\text{Initial Investment}}{\text{Cash inflow per period}}$$

Problem 1 (PB Period with even cash flow)

A firm is planning to undertake a project requiring investment of \$ 105 million. The project is expected to generate \$ 25 million per year for 7 years. Calculate PB period of the project

Solution

$$PB \text{ period} = \$105 \text{ million} / \$ 25 \text{ million} = 4.25 \text{ years}$$

After 4.25 years the firm will start earning profit.

2) PB Period for Uneven CF

When recovery or cash inflow are uneven then there is need to calculate the cumulative net cash flow for each period and we use the following formula:

$$PB = A + \frac{B}{C} \dots\dots\dots (1)$$

A = is the last period with -ve cumulative cash in flow and max re-payment

B = is the absolute remaining value of cumulative cash flow at the end of period A.

C = the total cash flow during the period after A.

Abbreviations

CF = Cash Flow

NCF = Net Cash Flow

CNCF= Cumulative Net Cash Flow

PB Period with Uneven CF

Problem PB Period with uneven cash flow

A firm is planning to undertake project that requiring initial investment of \$50 million and is expected to generate \$10 million in Year 1, \$13 million in Year 2, \$15 million in year 3, \$19 million in Year 4 and \$22 million in Year 5.

Calculate the PB value of the project

Solution

CF in Million

Year	NCF in \$	CNCF in \$	Net Value
0	(50)	(50)	(50)
1	10	10	-40
2	13	23	-27
3 (A)	15	38	-12 (B)
4	19 (C)	57	+7
5	22	79	+29

CF: Cash Flow

$$\text{Payback Period} = 3 + \frac{12}{19} \text{ years}$$

$$\text{Payback Period} = 3.31 \text{ years}$$

PB Period for uneven cash flow

Problem:

A management has allocated \$ 56,126 as investment to procure machine that generates annual Cash Flow (CF). In response management received 2 alternates. Each machinery generates a Cash Flow (CF) over a period of 5 years as follows:

Year	NCF Alternate – A \$	NCF Alternate – B \$
	<56,126>	<56,126>
1	14,000	22,000
2	16,000	20,000
3	18,000	18,000
4	20,000	16,000
5	25,000	17,000

Calculate:

- PB period of Alt - A and Alt – B
- Do you recommend Alt – A or Alt – B.

Solution

Year	NCF Alternate- A	Cumulat ive Net Cash Flow (CNCF) Alternat e-A	Net Cash Flow (NCF) Alternate-A	Year	NCF Alternate- B	CNCF- B	Net Value
0	<56, 126>	<56, 126>	<56, 126>	0	<56,126>		
1	14,000	14,000	-42,126	1	22,000	22,000	-34126
2	16,000	30,000	-26,126	2(A)	20,000	42,000	-14126 (B)
3(A)	18,000	48,000	-8,126 (B)	3	18,000 (C)	60,000	+ 3874
4	20,000 (C)	68,000	+16,000	4	16,000	76,000	+ 19874
5	25,000	93,000	17,000	5	17,000	93,000	+36874

Alternate – A

- Investment will be recovered 3 to 4 years.
- PB period will be 3 years plus fraction of 4th year.
- \$ 48,000 is recovered at the end of 3rd year i.e \$ 56,126 - \$ 48,000 = \$ 8,125 is yet to recovered
- In 4th year CF shows a return of \$ 20,000
- The fraction = $3 + \frac{8,126}{20,000}$
= 3 + 406 = 3.46 years

Alternate –B

- Investment will be recovered in 2nd or fraction of 3rd year.
- Amount recovered in 2 years \$ 42,000
- i.e \$ 56,126 - \$ 42,000 = \$ 1,425
- Balance 1,425 is to be recovered in 3rd year.
- In 3rd year the return is \$ 18,000
- This PB period = $2 + \frac{14,126}{18,000} = 2 + 0.785 = 2.785$ year

3) PB period with Discounted CF

PB Period with discounted Cash Flow takes into account time – value money

Problem 1.

An initial investment of \$2,324,000 is expected to generate \$600,000 per year for 6 years. Calculate the discounted payback period of the investment if the discount rate is 11%.

Solution

$$PV = F(1 + i\%)^{-n}, PV = (1 + i\%)^{-n} \times F$$

Year n (1)	CF in \$ (2)	PV factor $1/(1 + i)^n$, i is 11% (3)	Discounted CF in \$ Col 2 x Col 3 (4)	CNCF in \$ (5)	CNCF subtracted from Investment CF in \$ (6)
0	<2,324,000>	1.0000	<2,324,000>	2,324,000	\$ -2,324,000
1	600,000	0.9009	540,541	540,541	- 1,783,459
2	600,000	0.8116	486,973	1027514	- 1,296,486
3	600,000	0.7312	438,715	1466229	- 857,771
4	600,000	0.6587	395,239	1861466	- 462,533
5 (A)	600,000	0.5935	356,071	2217539	(B) - 106,462
6	600,000	0.5346	320,785 (C)	2538324	+ 214,323

$$PB = A + \frac{B}{C} \dots\dots\dots (1)$$

$$\text{Discounted Payback Period} = 5 + \frac{106,462}{320,785} = 5.32 \text{ years}$$

Service – Life Analysis

Service life

From Wikipedia, the free encyclopedia

A product's **service life** is its period of use in service.

It is mostly used in a non-technical and technical items.

Several other terms more accurately describe a product's life, from the point of manufacture, storage, and distribution, eventual use and repayment of product

Service life has been defined as "a product's total life in use from the point of sale to the point of discard" and distinguished from replacement life, the period after which the initial purchaser returns to the shop for a replacement repayment.

Determining a product's expected service life as part of business policy involves using tools and calculations from maintainability and reliability analysis

Service – Life Analysis is used to determine the number of years an asset must be retained and used to recover its initial cost with interest rate if initial cost, annual cash flow and SV is given

$$\text{Formula: } PB = -P + \sum_{J=1}^n (CF) + (P/F, i\% n) \dots \dots \dots (1)$$

$$\text{Functional Formula: } PB = -P + (P/A, i\%, n) + SV (P/F, i\%, n) \dots \dots \dots (2)$$

$$PB = -P + A \left[\frac{(1+i)^N - 1}{i(1+i)^N} \right] + \left[SV \frac{1}{(1+i)^N} \right] \dots \dots \dots (3)$$

Where

P = 1st Cost of Equipment, Service

SV = Salvage Value

A = Annuities (Equal amount over equal period) or uniform series

n = Time period

i % = Interest rate

J = Frequency of period (Duration)

CF = Cash Flow

Problem 1.

A Project Manager purchased Sewer Clearing Machine at a cost of \$ 18,000. Annual cash flow being \$ 3,000 with SV \$ 3,000. The interest rate is 15%. The service life lies between 14 years to 16 years determine actual service life.

PB 14 years

$$PB = -18,000 + 3,000 \left[\frac{(1 + 15\%)^{14} - 1}{15\% (1 + 15\%)^{14}} \right] + 3,000 \left[\frac{1}{(1 + 15\%)^{14}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(1 + 0.15)^{14} - 1}{0.15 (1 + 0.15)^{14}} \right] + 3,000 \left[\frac{1}{(1 + 0.15)^{14}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(1.15)^{14} - 1}{0.15(1.15)^{14}} \right] + 3,000 \left[\frac{1}{(1.15)^{14}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(7.075) - 1}{0.15(7.075)} \right] + 3,000 \left[\frac{1}{(7.075)} \right]$$

$$\begin{aligned} PB &= -18,000 + 3,000 \left[\frac{6.075}{0.15(7.075)} \right] + 3,000 \left[0.141 \right] \\ &= -18,000 + 3,000 \left[\frac{6.075}{1.061} \right] + 3,000 \left[0.141 \right] \end{aligned}$$

$$= -18,000 + 3,000 \left[5.725 \right] + 3,000 \left[0.141 \right]$$

$$= -18,000 + 14,772 + 423$$

$$PB = -18,000 + 17,598$$

$$PB = -402$$

PB 16 years

$$PB = -18,000 + 3,000 \left[\frac{(1 + 15\%)^{16} - 1}{15\% (1 + 15\%)^{16}} \right] + 3,000 \left[\frac{1}{(1 + 15\%)^{16}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(1 + 0.15)^{16} - 1}{0.15 (1 + 0.15)^{16}} \right] + 3,000 \left[\frac{1}{(1 + 0.15)^{16}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(1.15)^{16} - 1}{0.15 (1.15)^{16}} \right] + 3,000 \left[\frac{1}{(1.15)^{16}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(9.357) - 1}{0.15 (9.357)} \right] + 3,000 \left[\frac{1}{(9.357)} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{8.357}{1.403} \right] + 3,000 [0.106]$$

$$PB = -18,000 + 3,000 (5.956) + 3,000 (0.106)$$

$$PB = -18,000 + 17,868 + 318$$

$$PB = 186$$

PB 15 years

$$PB = -18,000 + 3,000 \left[\frac{(1 + 15\%)^{15} - 1}{15\% (1 + 15\%)^{15}} \right] + 3,000 \left[\frac{1}{(1 + 15\%)^{15}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(1 + 0.15)^{15} - 1}{0.15 (1 + 0.15)^{15}} \right] + 3,000 \left[\frac{1}{(1 + 0.15)^{15}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(1.15)^{15} - 1}{0.15 (1.15)^{15}} \right] + 3,000 \left[\frac{1}{(1.15)^{15}} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{(8.137) - 1}{0.15 (8.137)} \right] + 3,000 \left[\frac{1}{(8.137)} \right]$$

$$PB = -18,000 + 3,000 \left[\frac{7.137}{1.220} \right] + 3,000 [0.122]$$

$$PB = -18,000 + 3,000 (5.85) + 366$$

$$PB = -18,000 + 17,550 + 366$$

$$PB = -84$$

Interpolation

Divide 18,000 by 15 = 1,200 / year

Divide 1,200 by 12 = 100 / month

Divide 100 by 30 = 3.33 / day

Divide 84 by 3.33 = 25.2 say 26 day

i.e. PB = 15 year and 1 month